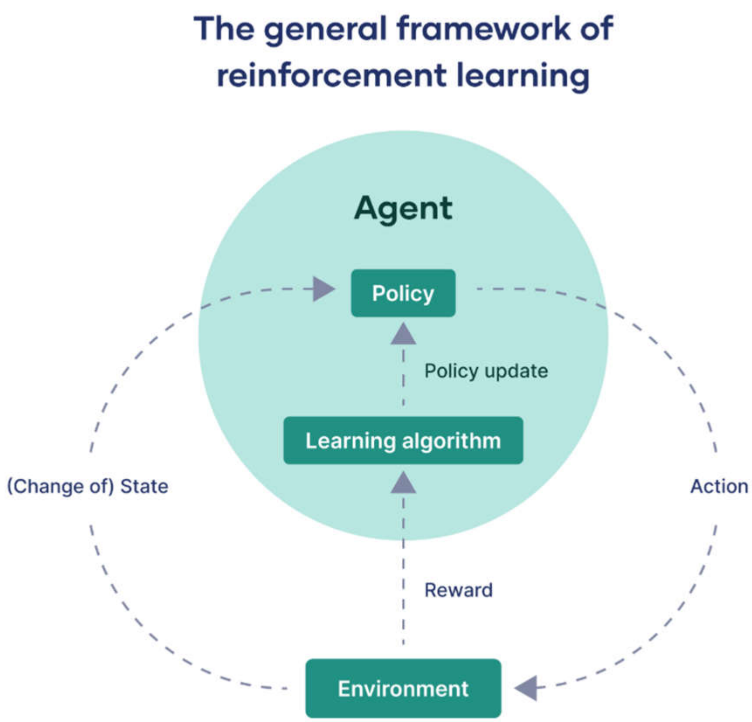
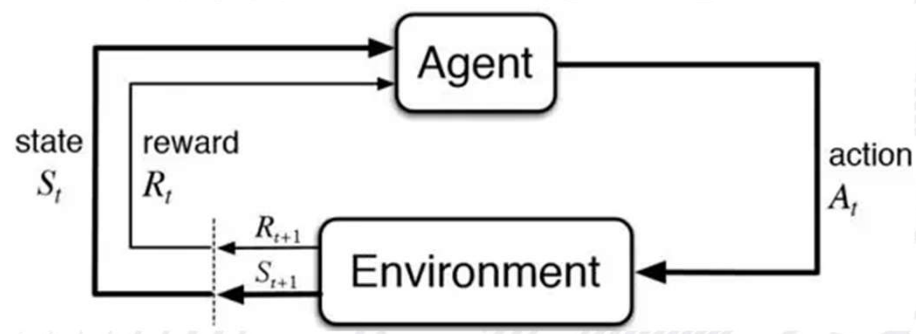
Chapter 4
Reinforcement learning

Q-Learning

Q-learning is a value-based reinforcement learning algorithm. The goal of Q-learning is to learn the optimal action-selection policy for an agent interacting with an environment.





- The agent learns by updating a table of values — called the Q-table — where each entry represents the value of taking a particular action in a given state.
- The Q-learning algorithm uses the following formula to update the Q-value for a state-action pair

$$Q(s,a) = Q(s,a) + \alpha\left(R + \gamma maxQ(s',a') - Q(s,a)\right)$$

  - Q(s, a) is the Q-value for state s and action a.
  - alpha is the learning rate
  - R is the immediate reward for taking action a in state s.
  - maxQ(s', a') is the maximum Q-value for the next state s', representing the best possible reward achievable from that state.

Old Q Value Reward Discount Rate (0 ~ 1)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\underbrace{(r + \gamma \max_{a'} Q(s',a') - Q(s,a))}_{\text{TD error}}$$

New Q Value

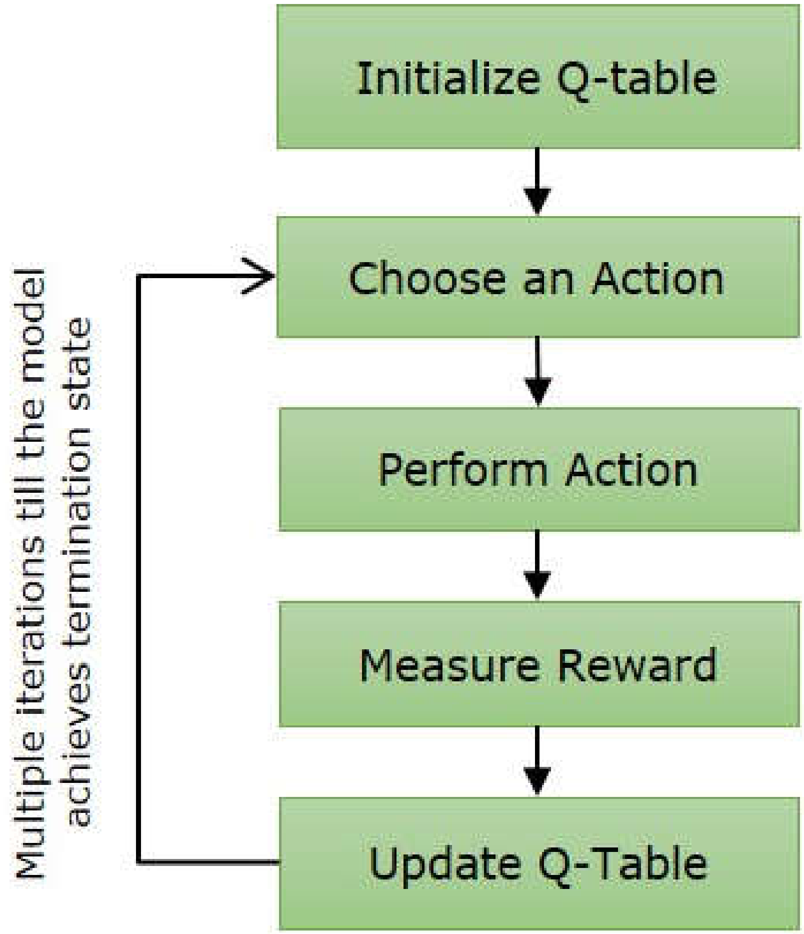
Learning Rate (0 ~ 1)

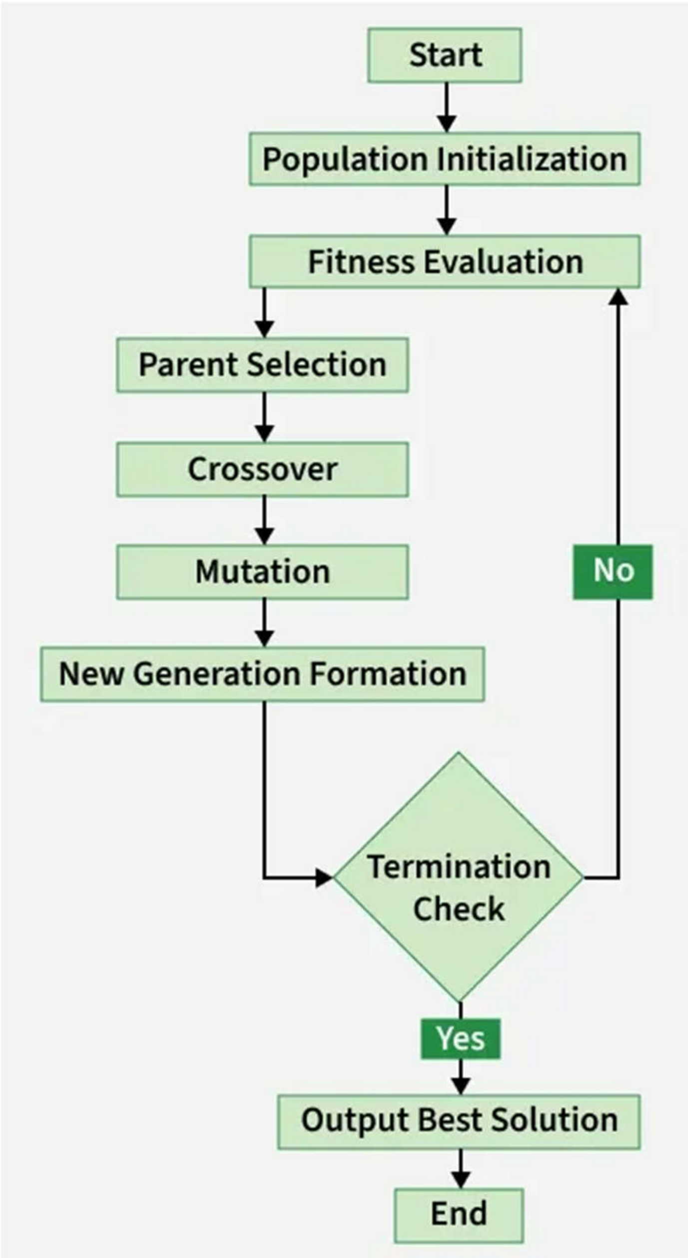
Maximum Q value of transition destination state

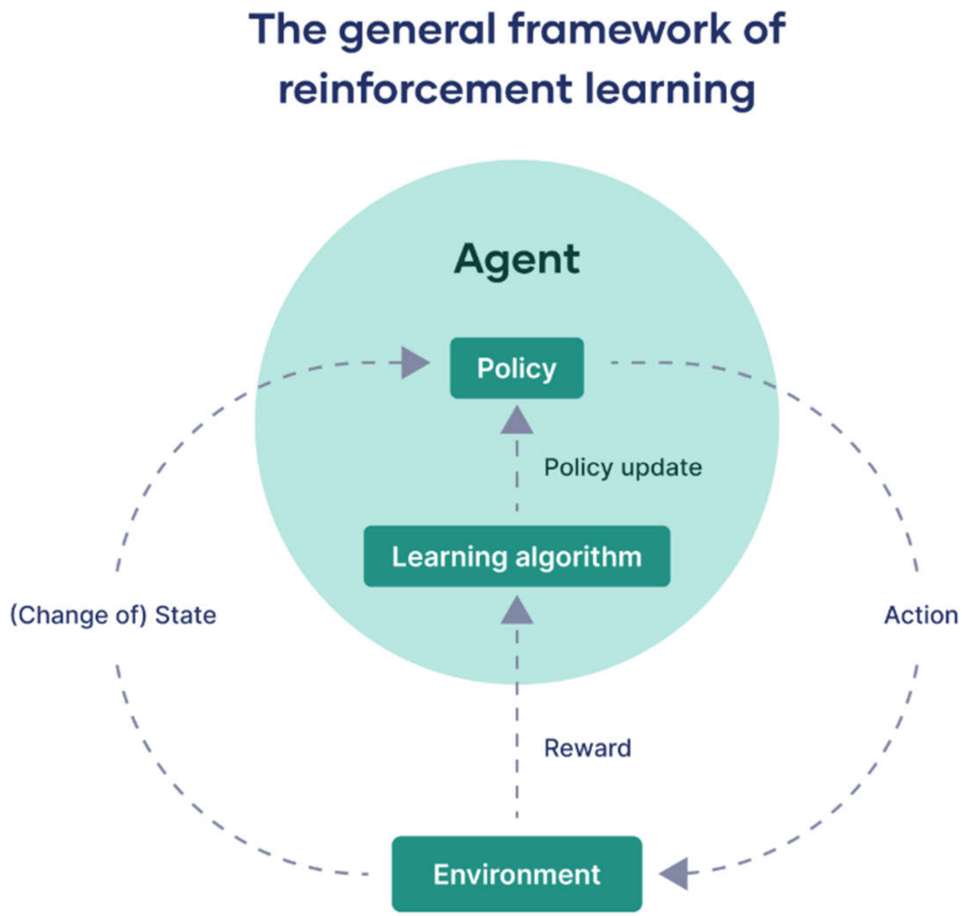
TD error

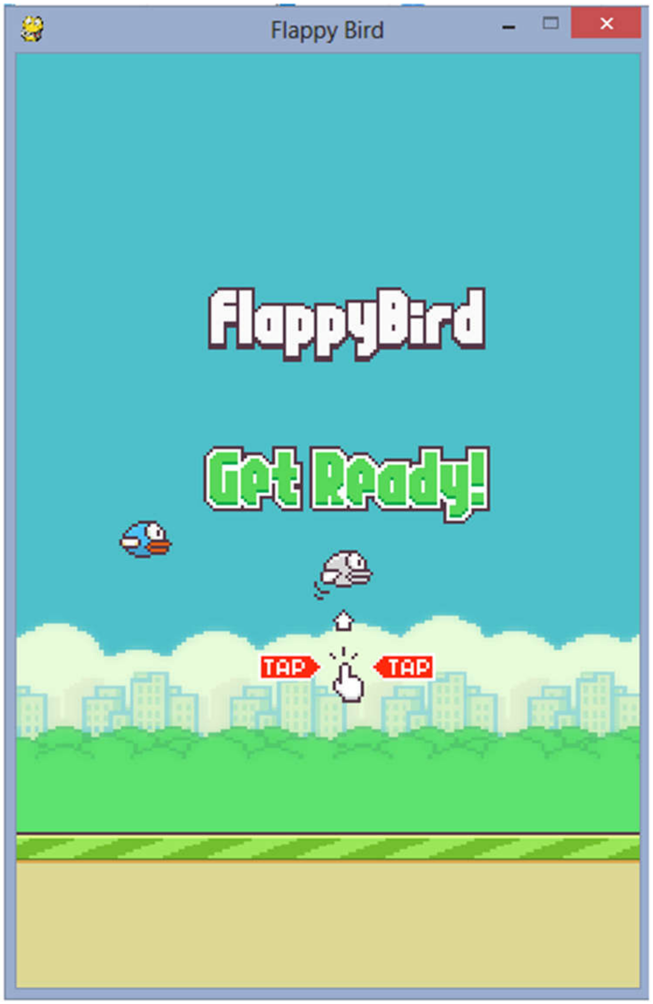
TD: Temporal Difference

Q-Learning Algorithm

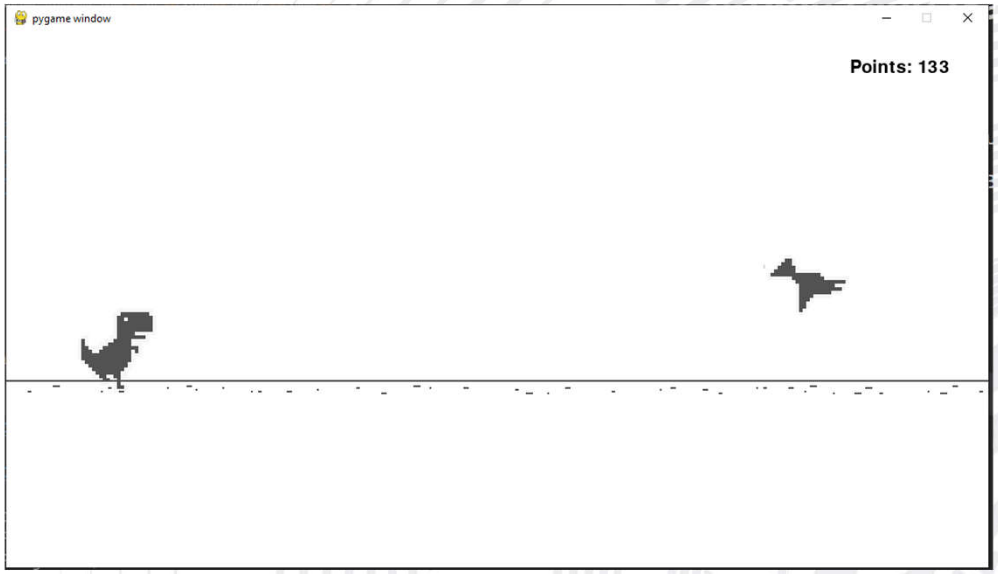



Start
Population Initialization
Fitness Evaluation
Parent Selection
Crossover
Mutation
No
New Generation Formation
Termination Check
Yes
Output Best Solution
End


The general framework of reinforcement learning
Agent
Policy
Policy update
Learning algorithm
(Change of) State
Action
Reward
Environment


Flappy Bird
FlappyBird
Get Ready!
TAP
TAP


Snake Game
Python Beginner Tutorial


pygame window
Points: 133

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Thank you!

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